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REVIEWERS: ARMY - MO
AIR FORCE - 85
- 69

USERS: NAVY - YD
- MC

P.A. ARMY - MO(ERDL)
Other Cust.
NAVY - YD
AIR FORCE - 69

PROCUREMENT SPECIFICATION
MIL-S-3785

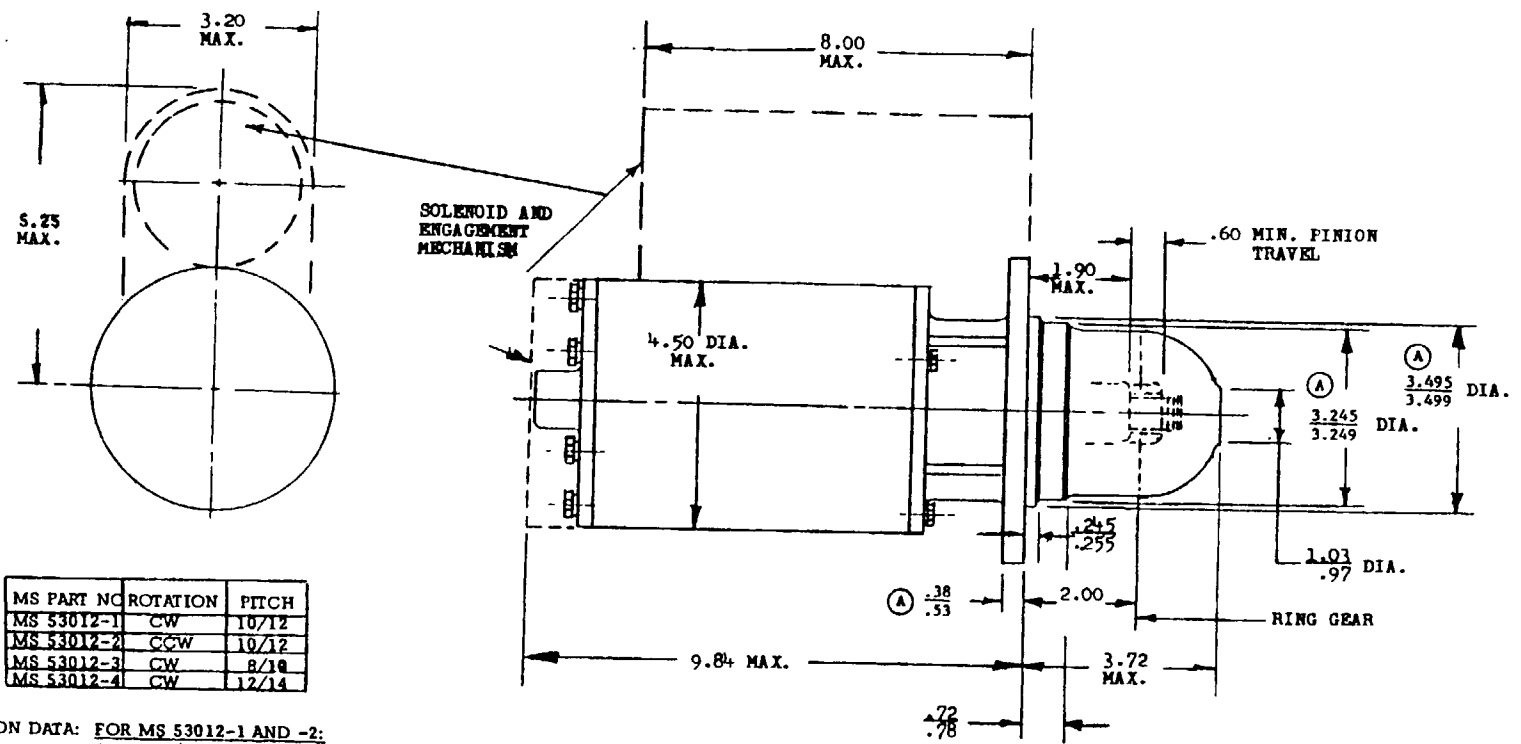
SUPERSEDES:

TITLE
STARTER, ENGINE, ELECTRICAL: 24-VOLT, D.C., 4-1/2-INCH FRAME DIAMETER, SOLENOID-ACTUATED, LIGHT-DUTY

MILITARY STANDARD
MS 53012

SHEET 1 OF 4

DD FORM 672-1 (Coordinated)
PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.



Ⓐ

MS PART NO	ROTATION	PITCH
MS 53012-1	CW	10/12
MS 53012-2	CCW	10/12
MS 53012-3	CW	8/10
MS 53012-4	CW	12/14

NOTES:

Ⓐ 1. PINION DATA: FOR MS 53012-1 AND -2: 9 TEETH (10-TOOTH BLANK); 10/12 PITCH; .900 P.D.; 1.167 O.D.; 20° PRESSURE ANGLE; .015 TO .025 P.L. CLEARANCE. FOR CENTER DISTANCE LAYOUT USE 1.000 P.D.
FOR MS 53012-3: 9 TEETH (10-TOOTH BLANK); 8/10 PITCH; 1.125 P.D.; 1.450 O.D.; 20° PRESSURE ANGLE; .015 TO .025 P.L. CLEARANCE.
FOR MS 53012-4: 9 TEETH (10-TOOTH BLANK); 12/14 PITCH; .750 P.D.; 1.0164 O.D.; 12° PRESSURE ANGLE; NO PITCH LINE CLEARANCE.

Ⓐ 2. MOTOR MAY BE USED WITH EITHER NEGATIVE OR POSITIVE GROUNDED SYSTEM.

3. SPECIAL FLANGE MOUNTING.

Ⓐ 4. TERMINAL THREAD SIZE: BATTERY - 3/8-16 UNC 2A. SWITCH - #10-32 UNF 2A.

5. DESIGN AND ARRANGEMENT OF SOLENOID AND ENGAGEMENT MECHANISM ASSEMBLY ARE OPTIONAL, PROVIDED SHIFT LEVER AND LINKAGE ARE INCLOSED AND ALSO PROVIDED TERMINAL THREAD SIZES CONFORM TO AND LIMITING OVERALL DIMENSIONS DO NOT EXCEED THOSE SPECIFIED HEREIN.

6. ROTATION VIEWING DRIVE END.

Ⓐ 7. TOLERANCE: DIMENSIONAL: ± .030 UNLESS OTHERWISE SPECIFIED. ANGULAR: ± 1°.

8. THIS MOTOR IS IDENTIFIED AS TYPE II (RING GEAR ENGAGEMENT), GRADE A (WATERPROOF), CLASS 1 (POSITIVE DRIVE), OF REFERENCED SPECIFICATION.

9. MOTOR SHALL BE EQUIPPED WITH OIL SEAL FOR WET CLUTCH OPERATION.

(NOTES CONTINUED ON SHEET 2)

Ⓐ FOR CHANGES SEE SHEETS 1 THRU 4

FED. SUP CLASS 2920

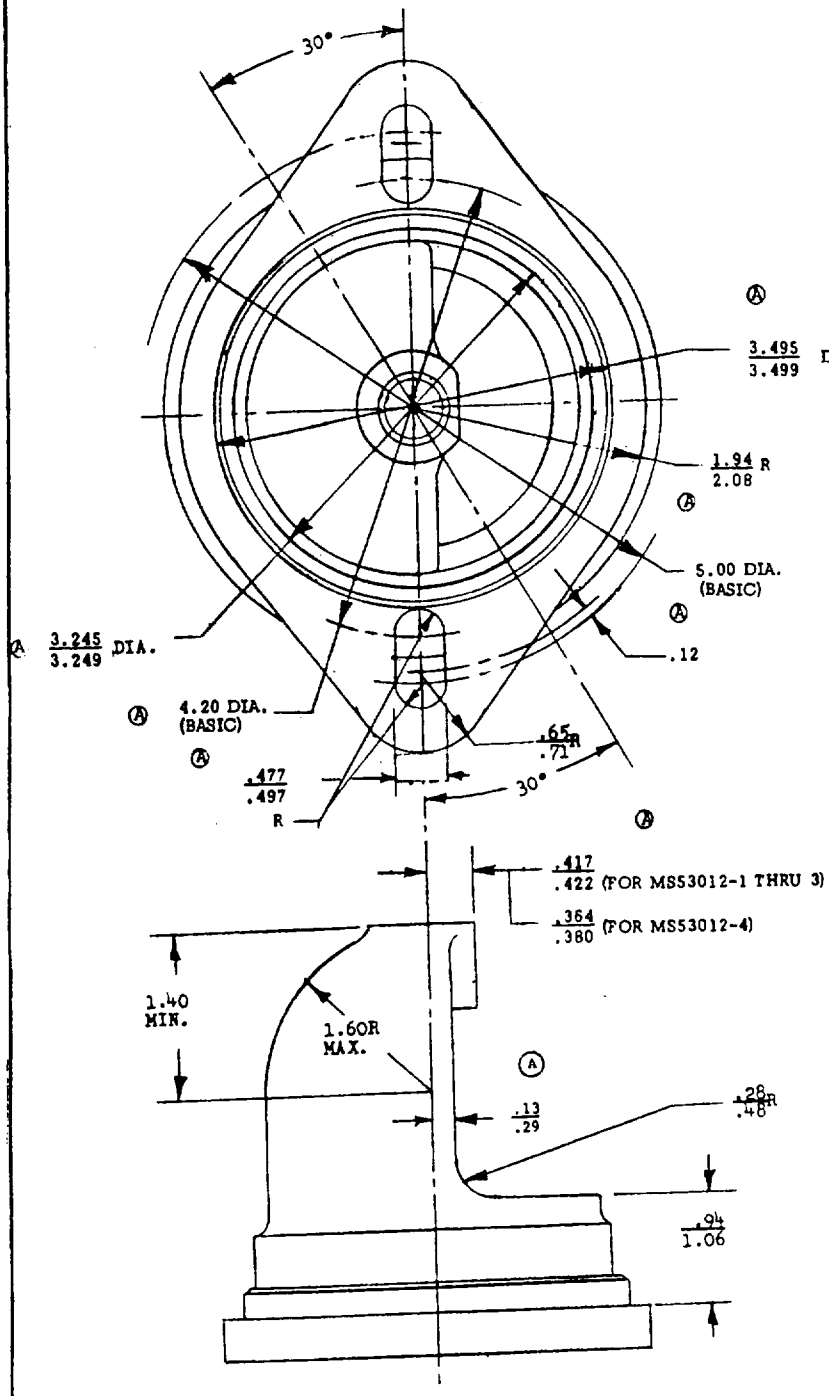
APPROVED 2 MAY 1962 REVISED Ⓐ 28 January 1965

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(CONTINUATION OF NOTES FROM SHEET 1)

10. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.
11. MODIFICATIONS IN OUTSIDE APPEARANCE AND CONSTRUCTION ARE ACCEPTABLE IF UNIT INTERCHANGEABILITY IS NOT AFFECTED.
12. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
13. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

APPROVED 2 MAY 1962 REVISED FOR CHANGES SEE SHEETS 1 TH

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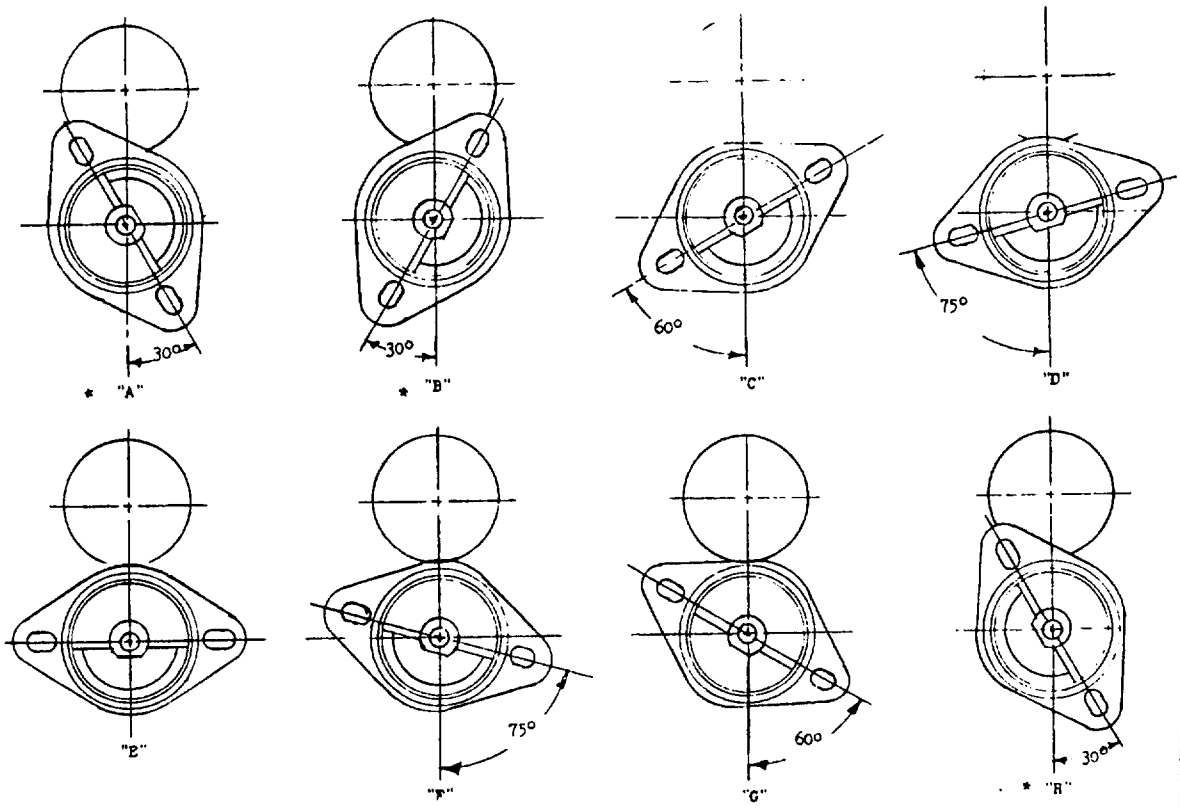
P.A. ARMY - MO(ERDL) Other Cust. NAVY - YD AIR FORCE - 69	TITLE STARTER, ENGINE, ELECTRICAL: 24-VOLT, D.C., 4-1/2-INCH FRAME DIAMETER, SOLENOID-ACTUATED, LIGHT-DUTY	MILITARY STANDARD MS 53012
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 2

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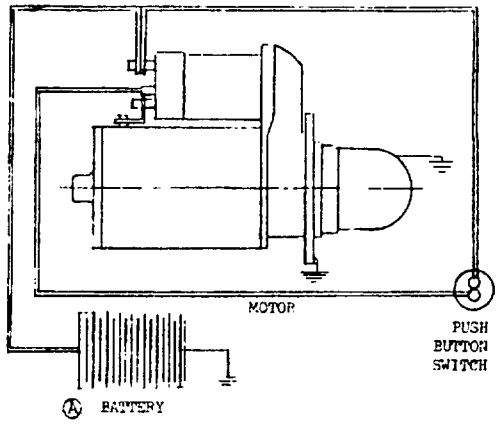
REVIEWERS: ARMY - MO
AIR FORCE - 85
- 69
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- MC



Ⓐ MOUNTING FLANGE AND PINION HOUSING SHALL BE ADJUSTABLE TO ALL POSITIONS OR PROVIDE EQUIVALENT POSITIONS SHOWN HEREON. MOUNTING POSITIONS VIEWING DRIVE END OF MOTOR.

A * POSITIONS FOR 5.00 DIA. BOLT CIRCLE ONLY

Ⓐ FIGURES "J," "K," "L," AND "M" HAVE BEEN DELETED



TYPICAL WIRING DIAGRAM

APPROVED 2 MAY 1962 REVISED Ⓐ FOR CHANGES SEE SHEETS 1 THRU 4

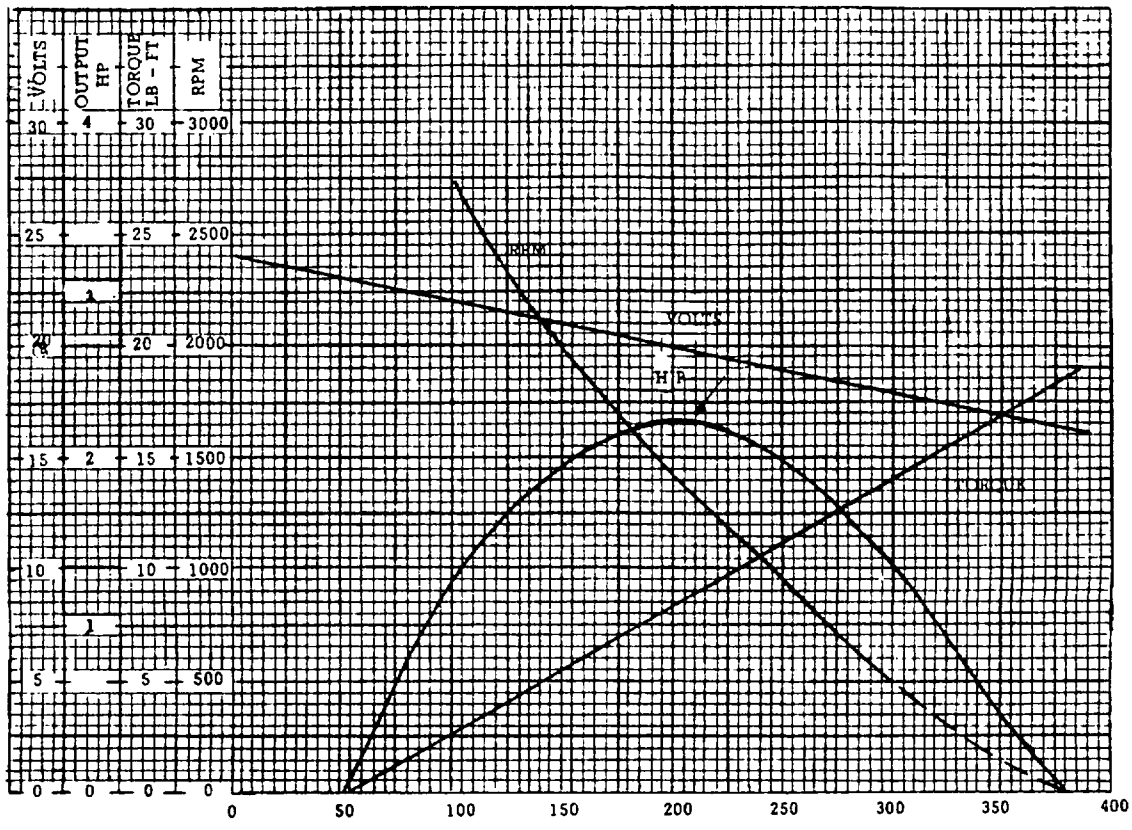
Defense standardization mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

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PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 3

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A M P E R E S

Ⓐ PERFORMANCE CURVE

(CONTINUATION OF NOTES FROM SHEET 2)

14. TORQUE AND R.P.M. MUST NOT BE LESS THAN 10% BELOW THAT SHOWN BY CURVES, WITH VOLTAGE APPLIED AS SPECIFIED. OUTPUT CURVE IS NOT RELATED TO TORQUE AND R.P.M. CURVES SHOWN, BUT IS THE MINIMUM FOR ANY PARTICULAR MOTOR.

15. THE VOLTAGE CURVE SHOWN IS THE SAE STANDARD DUTY 24-VOLT BATTERY CURVE.

Ⓐ 16. LOCATION OF NAMEPLATE SHALL BE OPTIONAL. NAMEPLATE SHALL BE FASTENED TO STARTER FRAME WITH RIVETS OR SCREWS AND LOCK WASHERS.

Ⓐ 17. STALLED TORQUE DATA:

VOLTS- 12.0 (APPROX.)
CURRENT (AMPS) (MAX.) - 250
TORQUE (LB-FT) (MIN) - 12

Ⓐ 18. SOLENOID CURRENT CONSUMPTION (PULL-IN PLUS HOLD-IN) SHALL NOT EXCEED 29 AMPERES.

Defense and is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

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PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 4

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